SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 3.2: Linear Motion — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 3.2: Linear Motion
Driving Question	DRIVING QUESTION: Why does a runaway mkokoteni on the Limuru escarpment keep getting faster — and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon & Motion Vocabulary: Why Does the Mkokoteni Keep Getting Faster?**	Students watched/discussed the anchoring phenomenon: a mkokoteni (handcart) released on the steep Limuru escarpment road. They observed that the cart moves slowly at first, then progressively faster, and does not maintain a constant speed. Students noted qualitative observations (it speeds up, it travels further each second) before any vocabulary was introduced.	The five key terms of linear motion — distance, displacement, speed, velocity, and acceleration — and the critical scalar/vector distinction between them. Students learned that distance is the total path length (scalar), while displacement is the straight-line change in position with direction (vector); speed is the rate of change of distance (scalar), while velocity is the rate of change of displacement (vector); and acceleration is the rate of change of velocity. Formulae introduced: speed = distance/time; velocity = displacement/time.	The mkokoteni keeps getting faster because its velocity is changing — meaning it is accelerating. Because the escarpment applies a continuous unbalanced force (gravity component along the slope), the cart's velocity increases every second. Pedagogical note: Stress to students that 'faster' in everyday language maps precisely onto 'positive acceleration' in physics. Use the mkokoteni's downhill direction as the positive direction convention for all subsequent lessons to build consistency.
Lesson 2 **Equations of Linear Motion — Part 1: Deriving and Applying the Formulae**	Students examined velocity–time graphs drawn for the mkokoteni scenario — a straight line rising from an initial velocity u to a final velocity v over time t. They identified the gradient (acceleration, a) and the area under the graph (displacement, s). From the graph, students derived or were guided to derive the four equations of linear motion: (1) $v = u + at$, (2) $s = ut + \frac{1}{2}at^2$, (3) $v^2 = u^2 + 2as$, (4) $s = \frac{1}{2}(u + v)t$.	The four kinematic equations of linear motion apply exclusively to uniform (constant) acceleration in a straight line. Each equation links a specific combination of the five variables (u, v, a, s, t), and choosing the correct equation depends on which variable is unknown and which three are given. Students practised substituting mkokoteni values (e.g., $u = 0$, $a = 3$ m/s², $t = 5$ s) into each equation to build familiarity.	Because the mkokoteni accelerates uniformly down the Limuru escarpment slope, all four equations are valid tools for predicting its motion. The velocity–time graph being a straight line is direct graphical evidence of uniform acceleration. Pedagogical note: Emphasise that these equations are ONLY valid when acceleration is constant — a misconception students often carry forward. Return to this constraint explicitly when free fall is introduced in Lesson 4.

Lesson 3 **Equations of Linear Motion — Part 2: Problem Solving**	Students worked through structured mkokoteni-themed problems in which they practised a GIVEN–FIND–EQUATION–SOLVE–CHECK framework. Worked examples included: predicting the speed of the mkokoteni after 8 seconds from rest on a slope with $a = 3$ m/s²; calculating how far it travels before reaching a dangerous speed; and finding the time taken to travel a set distance along the Limuru road.	Students learned how to systematically identify GIVEN and FIND quantities before selecting an equation, reducing common algebraic errors. They learned that the four equations of linear motion can produce two solutions when solving quadratic forms (e.g., $s = ut + ½at^2$), and that the physically meaningful solution must be selected using context. Unit consistency (metres, seconds, m/s, m/s²) was reinforced as a problem-solving discipline.	Mathematical prediction of the mkokoteni's speed and position at any moment is now possible using the equations of uniform acceleration. For example, starting from rest ($u = 0$) with $a = 3$ m/s², after 10 s the cart reaches $v = 30$ m/s (≈ 108 km/h) — a dangerously high speed over a short distance. Pedagogical note: Cold-call at least 3 students to present their GIVEN–FIND steps aloud before solving; this surfaces equation-selection misconceptions early. Allow 2 minutes of wait time after posing each new problem type.
Lesson 4 **Investigating Free Fall: Does Mass Matter? Dropping Objects from the Limuru Escarpment Height**	Students conducted a controlled experiment dropping objects of different masses (e.g., a stone and a crumpled paper ball, or two stones of different sizes) from a fixed height representing the escarpment drop. They timed the fall using stopwatches and recorded height (h) and time (t) in a results table. Students also observed a video demonstration (or teacher demo) of a feather and coin falling in a vacuum to challenge the 'heavier falls faster' misconception.	Using $h = ½gt^2$, students calculated the value of g from their experimental data and found a class average close to 9.8 m/s² (accepted as 10 m/s² for calculations). This confirms that gravity gives ALL falling objects the same acceleration regardless of mass, when air resistance is negligible. Students also identified sources of experimental error (reaction time, air resistance on lighter objects) and suggested improvements.	If the mkokoteni were to fly off the escarpment edge (a thought-experiment extension), it would accelerate downward at $g = 10$ m/s² regardless of its loaded mass. This directly connects free fall to the anchoring phenomenon: the constant acceleration students measured in the lab is the same physical cause — gravity — that accelerates the mkokoteni down the slope. Pedagogical note: Ensure students record raw data and calculate g individually before sharing class averages; this reinforces SEP: Analysing and Interpreting Data.
Lesson 5 **Freely Falling Bodies — Explain & Calculate: Using g to Predict the Mkokoteni's Speed and Distance**	Students applied the free-fall simplified equations to a series of Kenyan-context problems: e.g., a mango dropped from a tree on a Kisumu farm; a stone thrown upward by a child at Uhuru Park; and an object falling from the top of the Limuru escarpment cliff. They plotted velocity–time and displacement–time graphs for each scenario and compared the shape (linear v–t graph; parabolic s–t graph).	When a body is released from rest ($u = 0$) under gravity ($a = g = 10$ m/s²), its motion is fully described by three simplified equations: $v = gt$ (speed increases linearly with time), $h = ½gt^2$ (distance fallen increases as the square of time), and $v^2 = 2gh$ (speed–distance relationship without time). Students learned that upward motion uses $a = –g$, and that at maximum height, $v = 0$.	The parabolic shape of the s–t graph for free fall explains why the mkokoteni covers much greater distances in each successive second — the same principle applies on the slope. Because displacement grows as t^2, small increases in time produce disproportionately large increases in distance and speed, making the situation rapidly dangerous. Pedagogical note: Explicitly connect the parabolic s–t graph here back to the qualitative observation students made in Lesson 1 (the cart moves further each second). This DQB closure moment should be narrated aloud.
Lesson 6 **Applications, Road Safety & Real-Life Contexts of Linear Motion**	Students analysed road-safety data and stopping-distance problems contextualised along Kenyan highways (e.g., the Nairobi–Nakuru highway near Limuru, the Mombasa road). They investigated how braking	Braking distance is proportional to the square of initial speed ($s = u^2/2a$ when $v = 0$), meaning that doubling speed quadruples stopping distance and tripling speed increases stopping distance nine times.	A mkokoteni or matatu travelling at high speed down the Limuru escarpment requires a disproportionately long distance to stop compared with one travelling slowly. This quadratic relationship is the

	distance changes with initial speed using the equation $v^2 = u^2 + 2as$ (with $v = 0$ and a = deceleration). A data table and graph of speed vs. stopping distance were constructed, revealing a non-linear (quadratic) relationship.	Students also learned the distinction between thinking distance (constant reaction time × speed, linear) and braking distance (quadratic), and combined them into total stopping distance. Road safety messaging — relevant to Kenya's high matatu accident rates — was linked directly to the physics.	mathematical reason why high-speed runaway vehicles are so difficult to control — a direct, quantitative answer to part of the driving question. Pedagogical note: Invite students to calculate stopping distances for 20 km/h, 40 km/h, and 80 km/h and share results with the class; use cold-calling to discuss why the NTSA enforces speed limits near escarpments.
Lesson 7 **Velocity–Time Graphs: Reading, Drawing, and Interpreting the Mkokoteni's Full Journey**	Students were given a mixed set of velocity–time graphs representing different phases of the mkokoteni's journey: initial rest, uniform acceleration down the slope, brief constant velocity on a flat section, and emergency deceleration to a stop. They practised reading gradient (acceleration/deceleration) and area under the graph (displacement) for each phase. Students also sketched their own v–t graphs from a written description of a matatu journey along the Nairobi–Limuru road.	The gradient of a v–t graph equals acceleration (positive gradient = acceleration; negative gradient = deceleration; zero gradient = constant velocity). The area under a v–t graph — calculated as triangles and rectangles — equals displacement. A horizontal line on a v–t graph does NOT mean the object is stationary; it means constant velocity. Students learned to extract all five kinematic variables from a single v–t graph, connecting graphical and algebraic methods.	The complete v–t graph of the mkokoteni's journey provides a visual, integrated answer to the driving question: the rising straight line during the escarpment phase shows constant acceleration; the large area under that line shows the large displacement covered. The graph makes it visually clear why the mkokoteni is so dangerous — it reaches high velocity quickly and covers great distance before any intervention. Pedagogical note: Allow pairs to peer-mark each other's sketched graphs using a provided mark scheme; this consolidates graphical literacy before the final lesson.
Lesson 8 **Consolidation, DQB Completion & Final Explanation — The Mkokoteni Mystery Solved**	Students returned to their Driving Question Boards, reviewed all questions posted across Lessons 1–7, and systematically answered each using evidence gathered throughout the unit. They constructed or revised their final cumulative model of the mkokoteni's motion — a labelled diagram with a v–t graph, key equations, and a written explanation — demonstrating full integration of all concepts. A class gallery walk allowed groups to compare and critique each other's models.	Gravity provides a constant unbalanced force on the mkokoteni, producing constant acceleration ($a \approx 3$ m/s^2 on the slope, or $g = 10$ m/s^2 in free fall). This means speed increases linearly with time ($v = u + at$), distance increases quadratically with time ($s = ut + \frac{1}{2}at^2$), and the stopping distance grows as the square of speed — making the runaway cart increasingly dangerous and increasingly hard to stop with every passing second. All four equations of linear motion, free-fall principles, and graphical analysis are tools that together give a complete, quantitative answer to the driving question.	The mkokoteni keeps getting faster because gravity exerts a continuous unbalanced force along the slope, producing constant acceleration. Using the equations of linear motion, we can predict exactly: (1) its speed at any time using $v = u + at$; (2) the distance it has travelled using $s = ut + \frac{1}{2}at^2$; and (3) how long it would take to stop using $v^2 = u^2 + 2as$. The mathematics transforms a dangerous real-world event into a predictable, quantifiable phenomenon — the core promise of physics. Pedagogical note: Close the unit by asking students to write one sentence answering the driving question in their own words; cold-call 4–5 students. Celebrate specific vocabulary use (acceleration, uniform, displacement, deceleration) as evidence of genuine learning.